

AppImage: A Promethean Idea

AppImages are distro-agnostic and can run in all available distros. They need not be installed in the system. Users can carry these with them, and start working in a completely new system. The concept could be game changing for developers. Let's see how it works.

As a novice Linux user, I choose a distro based on its architecture, stability, and the apps it can support. Among these three, I feel apps are an important factor because most Linux users are dependent on their distro's app store or its repositories for their app needs. This is where Linux falls behind the Windows OS.

Generally, Windows users download apps directly from their respective websites, which provide the most recent version of these apps. Also, apps are downloaded as a single package and users can proceed with the installation right away. But in Linux, the approach is slightly different. The user goes to the distro app store or installs the app via the console by issuing a command. Each distro can have varying versions of the same app. So it is safe to say that there's a lack of uniformity within the distros. To receive the most recent version of the software is not possible via the distro's app store or its repositories, unless your distro is a rolling/semi-rolling release. Sometimes, users may face some issues like "dependencies are not matching" or "dependencies are missing". In this case, users may struggle a lot because they have to find out the matching versions of dependencies, and install those versions in the system to finish the installation process.

But what if you can download a recent version of an app and, without installing it, start using it right away?

AppImage is a kind of app packaging system, which eliminates many obstacles



in the traditional Linux app packaging. In this article, we will briefly look at this system and how it works.

What's wrong with the traditional packaging?

Generally, the developer creates an app, and then releases it to the public. For Linux users, the developer needs to follow certain guidelines that are mandated by the various distros, so that the app can be incorporated into that distro's app store. Since Linux is diverse and developers have to make their apps available to all users, they need to repackage the app for other distros.

Except for the rolling release OSs, many of the Linux OSs are derived from the stable distros. Users are restricted to use the app version that is provided by the original distro, mainly because the OS maintainers focus on the security and stability of the OS.

But, sometimes, users wish to use a recent version of the app. This could be difficult if they rely on the distro's app store, because some distros can

take months to provide updates for the apps. Also, when you install an app via the traditional package manager it will look for the dependencies required and the compatibility of the existing dependencies. And if any discrepancies are found, it will find the required packages and ask you to install those too. But this dependency resolution is specifically made for the particular system that you are working on. So, typically, when you want to install the same app in another system, you will need to repeat the whole process.

This situation gets more complicated if the distro is unable to find the dependencies for that app. In that case, you will have to find the compatible version of the dependencies (sometimes in a tarball, in which case, you will need to build these too) and install them separately into the system.

What is an AppImage?

An AppImage is an app bundle that contains the app in a binary form along with its dependencies in a single file.

